



Day 8 Task - Browser, Email & Communication Forensics



Trios cyber • 11 Sept (Edited 14 Sept)

90/100 | Due 12 Sept

Practical Assignment: Browser, Email & Communication Forensics

Dear Students,

For this practical assignment, you are required to create controlled browser and email/communication activity on your own authorized computer and examine the resulting forensic artifacts.

The objective of this exercise is to understand how browser, email and communication artifacts can provide forensic information about web activity, searches, downloads, messages, attachments, timestamps, routing information and user activity.

Part 1: Browser Tool Installation & Setup

Install and configure a web browser that will be used to create and examine test activity.

Select ONE of the following:

- Google Chrome
- Mozilla Firefox

Obtain the browser from one of the official websites below:

- Google Chrome: <https://www.google.com/chrome/>
- Mozilla Firefox: <https://www.mozilla.org/firefox/>

Install the selected browser on your Windows desktop.

Create a separate test browser profile if the browser supports it.

Do not use sensitive personal browsing information for this practical.

After installation, record:

- Browser name
- Browser version
- Installation date



- Operating system
- Browser profile used for the practical

Take screenshots showing:

- Browser installed successfully
- Browser version
- Browser profile used for the practical

Part 2: Create Browser Activity

Using the test browser profile, create controlled browsing activity.

Perform activities such as:

- Visit at least 5 different websites
- Perform at least 3 different searches
- Download at least 2 test files
- Open several pages related to the searches
- Visit one website more than once
- Create at least one bookmark
- Open several tabs
- Close and reopen the browser

Use only safe and authorized websites and test files.

Record:

- Date/time of the activity
- Website visited
- Search performed
- File downloaded
- Download location
- Other relevant activity

Take clear screenshots of selected browser activity.

Part 3: Examine Browser Artifacts

Examine the browser artifacts created during the practical.

Examine available artifacts such as:

- Browser history
- Search history
- Cookies
- Web storage
- Browser cache
- Download history



- Saved sessions or recently opened tabs
- Bookmarks, where relevant

If the browser stores forensic information in SQLite databases, use a suitable SQLite database viewer to examine the database files.

If a particular artifact is not available, mention “Not Available.”

For at least 5 browser artifacts, record:

- Artifact name
- Location/path
- Type of information provided
- Relevant timestamp
- What you observed
- Why it may be relevant to an investigation

Take clear screenshots of the artifacts examined.

Part 4: Email Tool Installation & Setup

Install and configure an email application for the practical.

Use Mozilla Thunderbird as the primary email application.

Obtain Thunderbird from the official website:

- Mozilla Thunderbird: <https://www.thunderbird.net/>

Install Thunderbird on your Windows desktop.

Use a test email account or another authorized test environment.

Do not use confidential or private email information.

After installation, record:

- Email application name
- Version
- Installation date
- Test account used
- Mailbox/profile location, where available

Take screenshots showing:

- Thunderbird installed successfully
- Thunderbird version
- Test account or mailbox
- Mailbox structure



Part 5: Create Email and Communication Activity

Using the test email environment, create controlled communication activity.

Create or exchange test messages that include:

- At least 5 test email messages
- Different sender and recipient combinations where available
- Different subject lines
- Different timestamps
- At least 1 reply
- At least 1 email containing an attachment
- At least 1 archived message
- At least 1 deleted message

Use only test information.

Record:

- Sender
- Recipient
- Subject
- Date/time
- Message status
- Attachment name, where applicable
- Mailbox location

Take screenshots of the relevant messages and mailbox activity.

Part 6: Examine Email Headers, Metadata and Mailbox Artifacts

Examine the email messages and available mailbox artifacts.

Review:

- Sender information
- Recipient information
- Subject
- Date/time
- Message headers
- Routing information
- Message identifiers, where available
- Attachments
- Mailbox folders
- Archived messages
- Deleted messages
- Other available metadata



For at least 5 relevant email artifacts, record:

- Artifact name
- Location/source
- Metadata available
- What you observed
- Why it may be relevant to an investigation

Do not assume that a header field proves the identity of a person. Interpret each field according to what it actually records.

Take clear screenshots of the headers, messages, attachments and mailbox artifacts examined.

Part 7: Browser and Email Activity Reconstruction

Compare the browser activity with the email activity created during the practical.

Identify at least 3 relationships between the artifacts.

For example:

- A browser download followed by an email containing the same file
- A website visited before an email was sent
- A search followed by communication related to the subject of the search
- Matching timestamps between browser and email activity
- A downloaded attachment appearing in related file-system activity

For each relationship, record:

- Browser artifact
- Email artifact
- Date/time
- Observed relationship
- Interpretation

Create a chronological timeline containing at least 10 events, where sufficient information is available.

Clearly distinguish between:

- Direct evidence/observation
- Inference or interpretation

Part 8: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. Browser information



4. Email application information
5. Tools and versions used
6. Browser activity created
7. Browser artifacts examined
8. Email/communication activity created
9. Email artifacts examined
10. Screenshots
11. Timeline
12. Browser and email correlation
13. Direct observations vs inferences
14. Conclusion

Each screenshot should have a short caption explaining what it demonstrates.

Submission Requirements

Submit ONE file only:

- PDF (.pdf)
- Microsoft Word document (.docx)

Important Notes

Use only your own computer, test accounts, test files and authorized activity for this practical.

Do not examine another person's private browser or email data.

Do not modify original evidence if you later use a forensic copy for examination.

If an artifact is unavailable, clearly state "Not Available."

Your conclusion must be based on the available evidence. Do not state an inference as a confirmed fact.

Expected Learning Outcomes

After completing this practical, you should be able to:

1. Identify common browser forensic artifacts.
2. Examine history, searches, cookies, cache and download activity.
3. Examine email headers, metadata and mailbox artifacts.
4. Identify attachment, archived and deleted communication artifacts.
5. Build a basic web and communication timeline.
6. Correlate browser and email evidence.
7. Distinguish direct observations from forensic inferences.
8. Present a defensible reconstruction of activity based on available evidence.

Submission: Upload your completed PDF/DOCX report to this Google Classroom assignment before the specified deadline. Late submissions are allowed only till after 48 hours from the time the task was posted but it may result in deduction of marks.

